

# THEO KEETILE

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## PROFESSIONAL SUMMARY

Mechatronic engineering graduate with an MSc in Data Science, working across embedded firmware and PCB design through to full-stack delivery and the statistical modelling that establishes whether a system performed. Delivered the client-facing layer of a roadside assistance platform now deployed live for a commercial client, alongside predictive modelling and dashboard work on insurance claims, public expenditure and global energy supply. Seeking graduate roles where both disciplines are useful.

## EDUCATION

### MSc Data Science - University of Leicester

Sep 2025 - Sep 2026

- **Relevant Modules:** Modelling Data, Statistics for Data Science, Introduction to Functional Data Analysis, Information Visualisation, Fundamentals of Data Science, Excel for Data Science. Currently averaging 75% across completed taught modules.
- **Key Achievement:** Selected for an industry-commissioned MSc project with AB Analytics Ltd, delivering a working AI-powered vehicle recovery platform against client requirements.

### BEng (Hons) Mechatronic Engineering, 2:1 - University of Manchester

Sep 2022 - Jun 2025

- **Relevant Modules:** C Programming, Embedded Systems Project, Mechatronic Analysis & Design, Control Systems I & II, Microcontroller Engineering I & II, Sensors & Instrumentation.
- **Key Achievement:** Placed in the top 10 of 50 teams in the Embedded Systems Project, taking an autonomous vehicle from concept to a working, tested prototype.

### A-Levels - Cambridge Assessment International Education

2021

Results: Computer Science A\*, Physics A\*, Mathematics A.

### IGCSEs - Cambridge Assessment International Education

2019

Results: Seven A\* grades across nine subjects, including Mathematics, Physics, Chemistry and Computer Science.

## PROJECTS

### AI-Powered Vehicle Recovery Platform - MSc Project, commissioned by AB Analytics Ltd

Jun 2026 - Sep 2026

- **End-to-End Delivery:** Contributed to a two-sided vehicle recovery platform built in Flutter and Dart on a Django and PostgreSQL backend, with dispatch driven by K-nearest neighbours and Hungarian algorithm matching.
- **Stakeholder Delivery:** Translated client requirements into technical specifications with the company supervisor, delivering pricing and fee logic, job history reporting and localisation in a four-person team.

### Autonomous Line-Following Buggy - University of Manchester

Mar 2024

- **Embedded Development:** Wrote STM32 firmware in C covering sensor sampling, the line-tracking control loop and motor drive, and designed the control PCB in Altium from schematic capture to manufacture.
- **Team Leadership:** Led integration and iterative tuning against a physical track, resolving subsystem conflicts that only appeared once the modules were combined.

### Automotive Risk & Repair Cost Modelling - University of Leicester

Feb 2026

- **Predictive Modelling:** Applied logistic regression, decision trees and K-means clustering in R (caret, rpart, randomForest) across 268,255 insurance claim records, evaluated with ROC and AUC.
- **Responsible AI & Ethics:** Assessed the socio-economic impact of the models against EDI frameworks, verifying that predictions did not unfairly penalise marginalised communities.

### Council Expenditure Analysis & Forecasting Dashboard - University of Leicester

Dec 2025

- **Data Preparation:** Cleaned 20 months of Birmingham City Council supplier payments covering 3.15 billion pounds of spend in Power Query, standardising beneficiary and category fields across roughly 88,000 corrections.
- **Insight & Forecasting:** Built an interactive dashboard with slicers and pivot analysis exposing directorate spend, supplier concentration and transaction bands, and forecast quarterly liabilities by exponential smoothing.

### Robot Localisation Using a Particle Filter - University of Manchester

May 2025

- **Probabilistic Estimation:** Implemented particle filter localisation in MATLAB and C++, estimating robot pose within a known map from noisy sensor and motion models, and published the work on GitHub.

## LEADERSHIP EXPERIENCE

### Senior Karate Instructor - Bakwadi Karate Academy, Botswana

2017 - Present

- **Team Leadership:** Coached a squad of 30 athletes to the Cape Town Open international tournament, securing 11 gold medals through structured training plans and performance analysis.
- **Standards & Auditing:** Invigilated gradings and conducted adherence audits, and now supports the academy remotely on administration and project work. 3rd Dan black belt, Japan Karate-Do Hayashi-Ha Shito Ryu Kai.

### Regional Representative - Botswana Student Union UK & Ireland

Sep 2023 - Sep 2024

- **Stakeholder Management:** Acted as primary liaison between the Manchester and Liverpool student body and the Botswana High Commission, addressing welfare and academic concerns at institutional level.
- **Event Management:** Coordinated a sports tournament for over 100 participants and organised career development webinars, managing logistics and scheduling.

## TECHNICAL SKILLS

**Programming & Query:** Python, R, SQL, C, C++, MATLAB, Dart, Git, PostgreSQL.

**Machine Learning & Statistics:** Regression, classification, decision trees, random forests, K-means clustering, functional data analysis, hypothesis testing, forecasting.

**Embedded & Hardware:** STM32 firmware, Altium PCB design, control systems, sensors and instrumentation, power electronics.

**Libraries & Tools:** pandas, NumPy, SciPy, scikit-learn, caret, tidyverse, Quarto, Django, Flutter, Jupyter.

**Visualisation & Reporting:** Tableau, Advanced Excel (Power Query, pivot tables, KPI dashboards), Microsoft Office.

**Interests:** Automotive restoration and engineering, karate, fitness and nutrition, freehand technical drawing.